

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY

(Not to be reproduced without permission of BIS or used as an Indian Standard)

भारतीय मानक मसौदा
रंग रोगन, वार्निश और संबंधित उत्पादों के नमूनाकरण और
परीक्षण पद्धतियाँ
भाग 4 प्रकाशीय परीक्षण
अनुभाग 3 प्रकाश स्थिरता परीक्षण
[IS 101 (Part 4/ Sec 3) का चौथा पुनरीक्षण]

Draft Indian Standard

**Methods of Sampling and Test For
Paints, Varnishes and Related Products
Part 4 Optical Test
Section 3 Light Fastness Test
[Fourth Revision of IS 101 (Part 4/ Sec 3)]**

ICS 87.040

Paint, Varnishes and Related Products Sectional
Committee, CHD 20

Last Date for Comments: As per the covering letter

Paints, Varnishes and Related Products Sectional Committee, CHD 20

FOREWORD

(Formal Clause will be added later)

IS 101 "Methods of test for ready mixed paints and enamel", initially, was published as a unified standard in 1950. The standard was subsequently revised in 1961, 1964 and 1987. During the third revision, recognizing the need for clarity, the committee decided to restructure it by splitting into various parts based on type of tests. These parts included tests on liquid paints (general and physical), chemical examination, film formation, optical assessments, and mechanical tests on paint film formation. Each part was further subdivided into sections, addressing specific tests within those categories.

Accordingly, this (Part 4/Sec 3) was published in 1988, by superseding 12 of IS 101 : 1964 as third revision. It is for determination of light fastness of paints for interior use by exposure to light from artificial sources under prescribed conditions. This standard (Part 4/Sec 3) is one of a series dealing with methods of optical test for paints, varnishes, and related products.

The other sections of this Indian Standard (Part 4) are:

Sec 1 Opacity (Hiding power)

Sec 2 Colour-visual comparison of colour of paints

Sec 4 Gloss - determination of gloss value at 20°, 60° and 85°

This fourth revision has been undertaken to align the product standard with current market demands, thereby enhancing its acceptance and relevance. The following major changes have been incorporated:

- a) A method for determining the Degree of Colour Change (ΔE) using a spectrophotometer has been incorporated.
- b) References to relevant Indian Standards have been updated wherever necessary.

In reporting the result of a test or analysis made in accordance with this standard, If the final value, observed or calculated, is to be rounded off, it shall be done in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'.

Draft Indian Standard

**METHODS OF SAMPLING AND TEST FOR
PAINTS, VARNISHES AND RELATED PRODUCTS**

PART 4 OPTICAL TESTS

Section 3 Light Fastness Test

(Third Revision)

1 SCOPE

This standard prescribes a method for the determination of light fastness of paints for interior use by exposure to light from artificial sources under prescribed conditions.

2 APPARATUS

2.1 Apparatus embodying either a xenon lamp or carbon arc may be used.

Note — In case of dispute, the xenon arc shall be used as referee method.

2.1.1 Test chamber — The test chamber shall consist essentially of a ventilated enclosure at the centre of which the source or radiation shall be situated. The test panels shall be held on a suitable rack arranged symmetrically around the source so that the irradiance falling on any part of the panels does not vary from the mean value by more than ± 10 percent. The panel rack may be rotates continuously around the radiation source to improve the even distribution of the light.

The enclosure shall be so constructed that any ozone produced by the radiation source does not come into contact with the test panels.

The conditions in the test chamber shall be as follows:

Air temperature	$(35 \pm 5) ^\circ\text{C}$
Black panel temperature	$(50 \pm 5) ^\circ\text{C}$
Relative humidity	(65 ± 5) percent

2.1.2 Radiation source — The radiation source shall consist of a suitable xenon lamp or carbon arc, together with an appropriate filter system which does not significantly alter the visible output of the arc. The irradiance on the panels shall be $(100 \pm 25) \text{ W/m}^2$ in the range 310 nm to 400 nm. The irradiance at wavelengths shorter than 310 nm shall not exceed 0.5 W/m^2 .

Note 1 — If there is doubt as to whether a particular type of equipment is in accordance with this standard, this shall be established by the user, by the manufacturer of the equipment or by an independent test authority.

Note 2 — In view of the short-term fluctuations of the output of the carbon arc, it is essential that the average for, the irradiance shall be taken over a period of not less than 5 h.

2.1.3 The following items are also required.

2.1.3.1 Cover sheet — of aluminium foil.

2.1.3.2 Colour matching cabinet

2.1.3.3 Calibration standards — For example, blue wool scale standard.

2.1.3.4 Cardboard — of thickness approximately 0.5 mm and of rigid quality for mounting the dyed wool calibration standards.

2.1.3.5 Geometric grey scale

3 PREPARATION AND COATING OF TEST PANEL

3.1 Unless otherwise agreed, use a panel of hard aluminium and coat it with an etch primer. The dimensions of the panel shall be suitable for the apparatus being used but shall not in any case be smaller than 60 mm × 40 mm.

3.2 Coat the panel with the product under test by the specified method and allow it to dry (or stove) and age in the specified manner and for the specified time. If normal drying conditioners are specified, these shall be interpreted as a temperature of $(27 \pm 2) ^\circ\text{C}$ and a relative humidity of (65 ± 5) percent with free circulation of air and no exposure to direct sunlight.

4 PROCEDURE

4.1 Cover half of the test panel and where appropriate, the calibration standard (2.1.3.3) with the cover sheet (2.1.3.1) and exposed it in the apparatus for the specified period.

This procedure advocates the use of the unexposed area adjacent to the exposed area for comparison, which is useful for checking the progress of the exposure. Comparisons, however, shall be based on contrast with a reference sample or other agreed standard.

4.2 At regular intervals remove the test panel from the apparatus, remove the cover sheet and observe the degree of colour change on the test panel and standard.

The comparison shall be carried out in the colour matching cabinet (2.1.3.2). If agreed, the colour change may be determined instrumentally as defined in 4.3.

Use may be made of the geometric grey scale (2.1.3.5), in recording the degree of colour change. If the test is to be continued, carefully replace the cover sheet in its original position.

4.3 Determination of Degree of Colour Change

4.3.1 The colour of the test panel shall be compared against the stipulated shade as given in IS 5 or any agreed standard sample between purchaser and supplier. The colour difference between a pair of painted panels shall be calculated in terms of ΔE (Delta E) by using CMC (2:1) colour difference equation. From spectral measurement, QC program calculate CIE colour specifications- L^* a^* b^* C^* h^* based on daylight illumination, as described in 4.3.2.

4.3.2 A reflectance spectrophotometer is used for reflectance measurement. Like Visual Assessment requirement, the sample should be sufficiently large enough to measure using Large Area of View (LAV). Small sized samples measured using Small Area of View (SAV) may not have high repeatability. Sample size should be minimum 2 × 2 inches. The spectral measurement should be carried out using a reflectance spectrophotometer having following features:

Repeatability – Short-Term	: Better than 0.15 ΔE_{CIELAB}
Inter-Instrument Agreement	: Better than 0.25 ΔE_{CIELAB}
Wavelength Range	: 400 nm to 700 nm (360 nm to 780 nm preferable)
Wavelength Increment	: 10 nm

4.4 Access the degree of light fastness of the test panel in terms of the specified or agreed requirements.

5 MONITORING

5.1 The irradiance is specified (2.1.2) but is liable to fall during operation owing to progressive ageing of the lamps and filters, inefficient cleaning or incorrect operation. The user shall, therefore, carry out regular and frequent checks on the output of the radiation source.

5.2 The checking procedure is not specified in this standard. The principles on which a suitable method may be based include the following:

- a) Measurement of the spectral energy distribution by means of a suitable spectrophotometer;
- b) Measurement of irradiance in wave bands using a photo-electric device with suitable filters;
- c) Chemical actinometry with suitable filters; and
- d) Measurement of actinic effect, for example, by means of blue wool standards or photosensitive polymers.

5.3 Should the check procedure indicate that the irradiance has fallen outside the specified limits, action shall be taken to restore the radiation to its original level. This may entail renewal of the lamp, cleaning or renewal of the filters or attention to the operating conditions.